Trust Index Notes

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Abstract

Working notes on the definition of AI-judge trustworthiness.

1 Notation

Table 1: Notation.

Symbol	Meaning
Items, agents, and benchmarks	
x	candidate artifact
y^*	gold / reference artifact
$\hat{y},\ \hat{y}(c)$	agent output on input c
A	set of agents / solvers
$a \in A$	an agent / solver
B	set of benchmarks
$b \in B$	a benchmark
Judges and targets	
AJ	set of AI judge configurations
$aj \in AJ$	AI judge configuration (model, prompt, K_{aj}, aggregation)
TJ	set of target judges
$tj \in TJ$	a target judge
$\theta_j,\ \theta_{aj}$	parameters of aj (prompt, weights, feedback, …)
$h_j,\ H$	human / target judge
t_j	target persona (or the target itself)
a_j	persona realized by configuration aj
$\pi,\ \Pi$	rater persona; set of personas
K_{aj}	number of judge repeats
K_a	number of agent generations
K_h	number of human passes
Scores and distributions	
$s_{aj}(x, y^*, \hat{y})$	score of aj on a triple; range $[0, 1]$
$\hat{s}_{aj}$	finite-sample estimate of s_{aj}
$s^*_T,\ \hat{s}^*_T$	target score and its finite-sample version
$p_{AJ}(s \mid \cdot)$	score distribution (pdf/pmf) of the AI judge
$p_T(s \mid \cdot)$	score distribution (pdf/pmf) of the target judge
$\hat{p}_{AJ},\ \hat{p}_T$	finite-sample versions of the above
Trust objects	
$\mathrm{TW}_m(aj)$	trustworthiness of aj at matching level m (construct)
$\mathrm{TI}_m(aj)$	Trust Index of aj at matching level m (criterion)
$\mathrm{PTI}_p(aj)$	Proxy Trust Index of aj on property set p (instrument)
$m,\ m_0, m_1, m_2$	matching-criterion index and its ladder
$\approx_\Pi$	persona precondition ($t_j = a_j$)
$\approx_m$	match under criterion m

Notation note. $\hat{p}_T$ denotes the target-judge pdf/pmf; the hat indicates that the estimate is finite-sample. We write $\hat{p}_T$ (or $\hat{p}_H$ when the target is human) and $\hat{p}_{AJ}$ for the AI-judge side.

2 Behaviors and agents

Definition 1 (Universally trustworthy AI judge). Let B be a set of benchmarks and A a set of agents. An AI judge $aj \in AJ$ is universally trustworthy if, for every benchmark $b \in B$ and every agent/solver $a \in A$, aj distributionally matches the corresponding target judge ($tj \in TJ$) (e.g., a human, a persona, a strict rubric, a lenient rubric, etc.):

$$\forall b \in B,\ \forall a \in A, \quad \text{if } t_j = a_j \quad \Rightarrow \quad \hat{p}_{AJ}\big(s \mid x,\, \hat{y}(c),\, y^*,\, \theta_j\big) \;=\; \hat{p}_T\big(s \mid x,\, \hat{y}(c),\, y^*,\, \theta_* + t_j\big),$$

where equality may be measured by KL divergence (or total variation).

Remark. The precondition $t_j = a_j$ (persona / target match) is required. If, for example, we set $t_j =$ strict but the judge a_j has parameters/configuration θ_{a_j} (prompt, weights, etc.) that map to a lenient judge, then the definition is unsatisfiable.

Remark (preference vs. trustworthiness). The literature typically uses "trustworthiness" implicitly to mean matching humans (human judgments; sometimes loosely "human preferences"). That is not RLHF preference (which of two answers to prefer). Preference can also denote the objective itself, so the two notions are close: preference is the more general term and can include trustworthiness. Once a frame of reference is fixed (objective or not—e.g. a persona), the distinction collapses; there is no residual gap between "I do not trust it" and "I do not prefer it."

3 Practical instantiations

Definition 2 (Common practice (m_1)). Most work treats an AI judge as trustworthy when its scores correlate with human scores on a panel:

$$\mathrm{TI}_{m_1}(aj) \;=\; \mathrm{Corr}\big(\hat{s}_{aj},\, \hat{s}_T\big).$$

This is a correlational match (dependence between the two score vectors), not a full distributional match (m_0) and not a pure first-moment match.

Definition 3 (Valid-and-stable (m_2 / $\mathrm{TI}_{2'}$)). A stricter practical criterion combines validity (correlation) with stability (low repeat variance) via a geometric mean:

$$\mathrm{TI}_{2'}(aj) \;=\; \Big(\mathrm{Corr}\big(\hat{s}_{aj},\, \hat{s}_T\big) \cdot \big(1 - \mathrm{std}(\hat{s}_{aj})\big)\Big)^{1/2}.$$

This is the working definition of trustworthiness$_{2'}$ used in these notes.

Remark (ideal vs. practical). The ideal target is full distributional match of scores once a persona is fixed (Definition 1). In practice this is expensive: many humans and many repeated scores would be required. We therefore approximate the ideal by matching a low-order moment (correlation of scores) and requiring the judge itself to be stable (low repeat standard deviation). These two quantities are combined by geometric mean to obtain $\mathrm{TI}_{2'}$. The practical criterion only captures limited moments of the ideal match.